

Food Delivery Time Prediction

A delivery person, a man with a beard wearing a tan jacket and dark pants, is riding a bicycle. He is carrying a large, yellow and silver insulated delivery bag on his back. The background is a dark, modern building with large glass windows. The overall tone is professional and focused on the delivery industry.

Data Science



OVERVIEW

Objectives, problems

METHODOLOGY

Methodolgy that we use

MODELING APPROACH

MODEL PERFORMANCE & INTERPRETATION

Machine Learning

CONCLUSION

Business recommendation

FOOD DELIVERY DATASET FROM KAGGLE

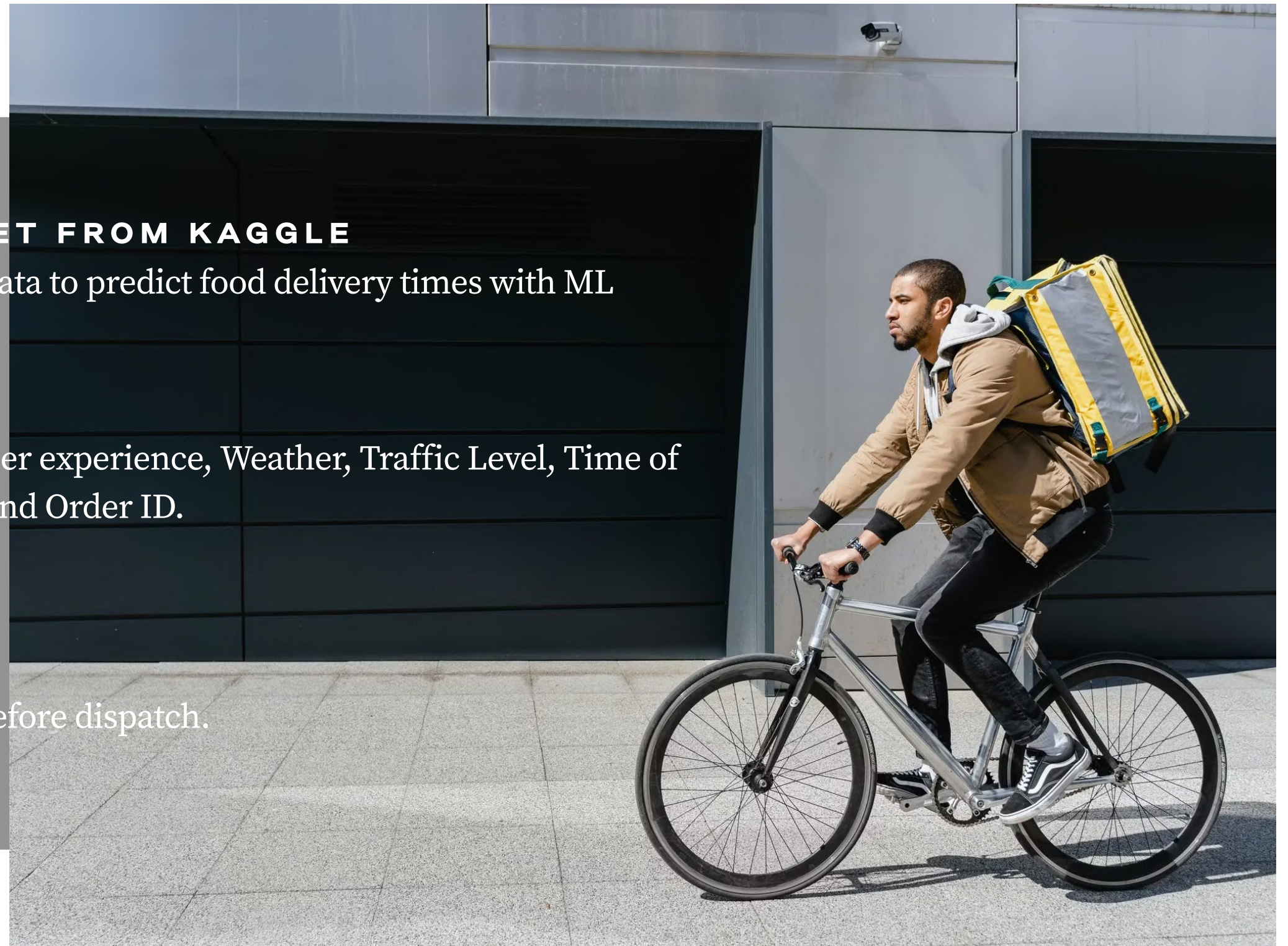
This dataset contains simulated data to predict food delivery times with ML

DATA

Distance, Preparation time, Courier experience, Weather, Traffic Level, Time of Day, Vehicle type, Delivery time and Order ID.
1000 rows, 9 columns.

GOAL

Predict delivery time (minutes) before dispatch.



METHODOLOGY



EDA

- Importing data
- Cleaning data: Drop Duplicates;
- Clean missing value ~30/1000
 - categorical = mode
 - numerical = median
- Visual

Feature Engineering

- One hot encoding categorical → numerical

Machine Learning

- Train/Test: 80/20, random_state=42
- Target & Metric: Predict Delivery_Time_min, optimize MAE (minutes)
- Models (baseline): Linear Regression, Decision Tree, Random Forest
- Tuned Models (GridSearchCV, 3-fold)

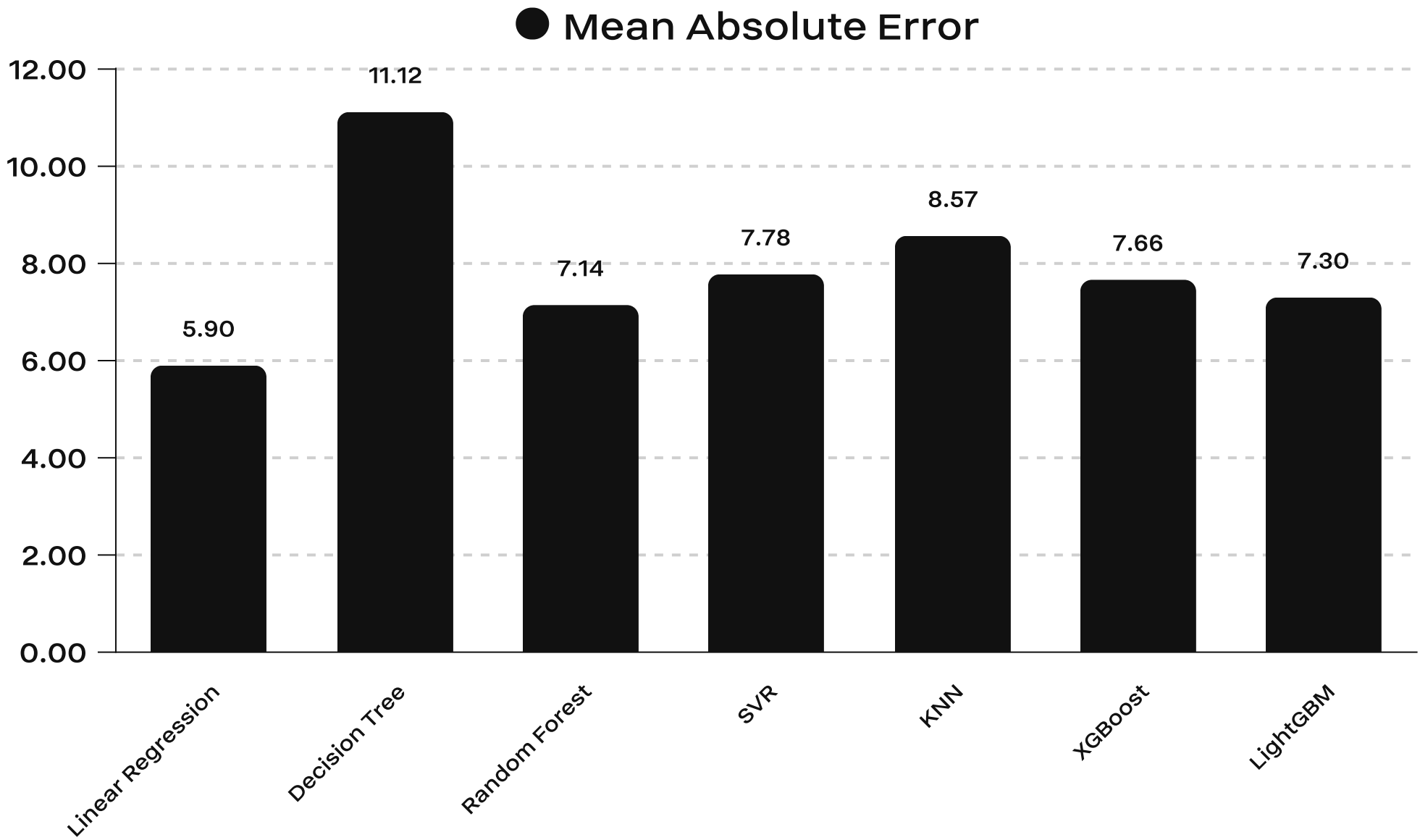
Business Recommendation

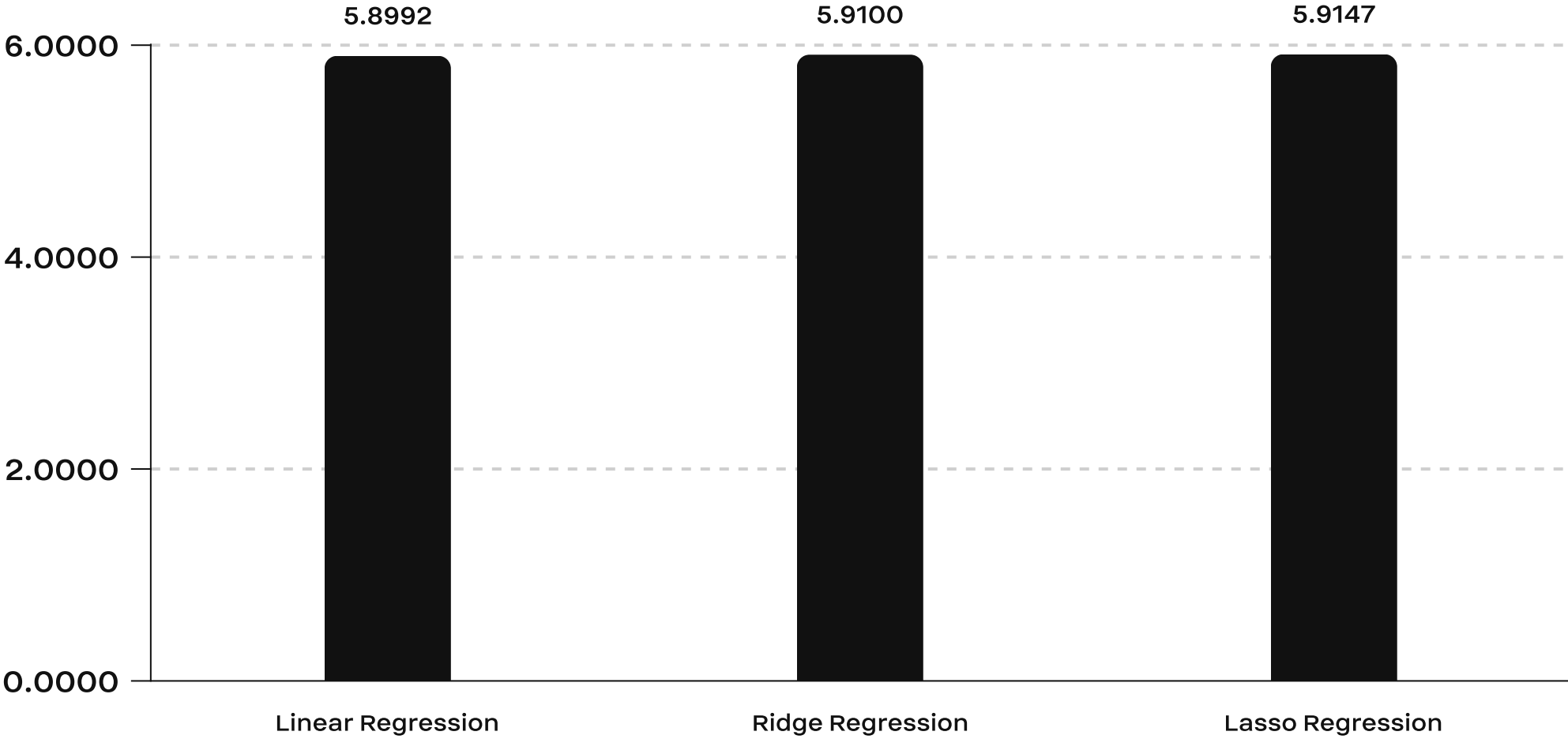
BASELINES

- Linear Regression, Decision Tree, Random Forest
- **KNN Regressor** (*key knobs: $n_neighbors$, $weights$*)
- **SVR (RBF)** (*key knobs: C , $epsilon$, $gamma$*)
- Tuned ensembles: Random Forest, LightGBM, XGBoost (GridSearchCV, 3-fold)
- Take four lowest for tuning

MAE (minutes) — lower is better

Best MAE: Linear Regression (5.90 min)

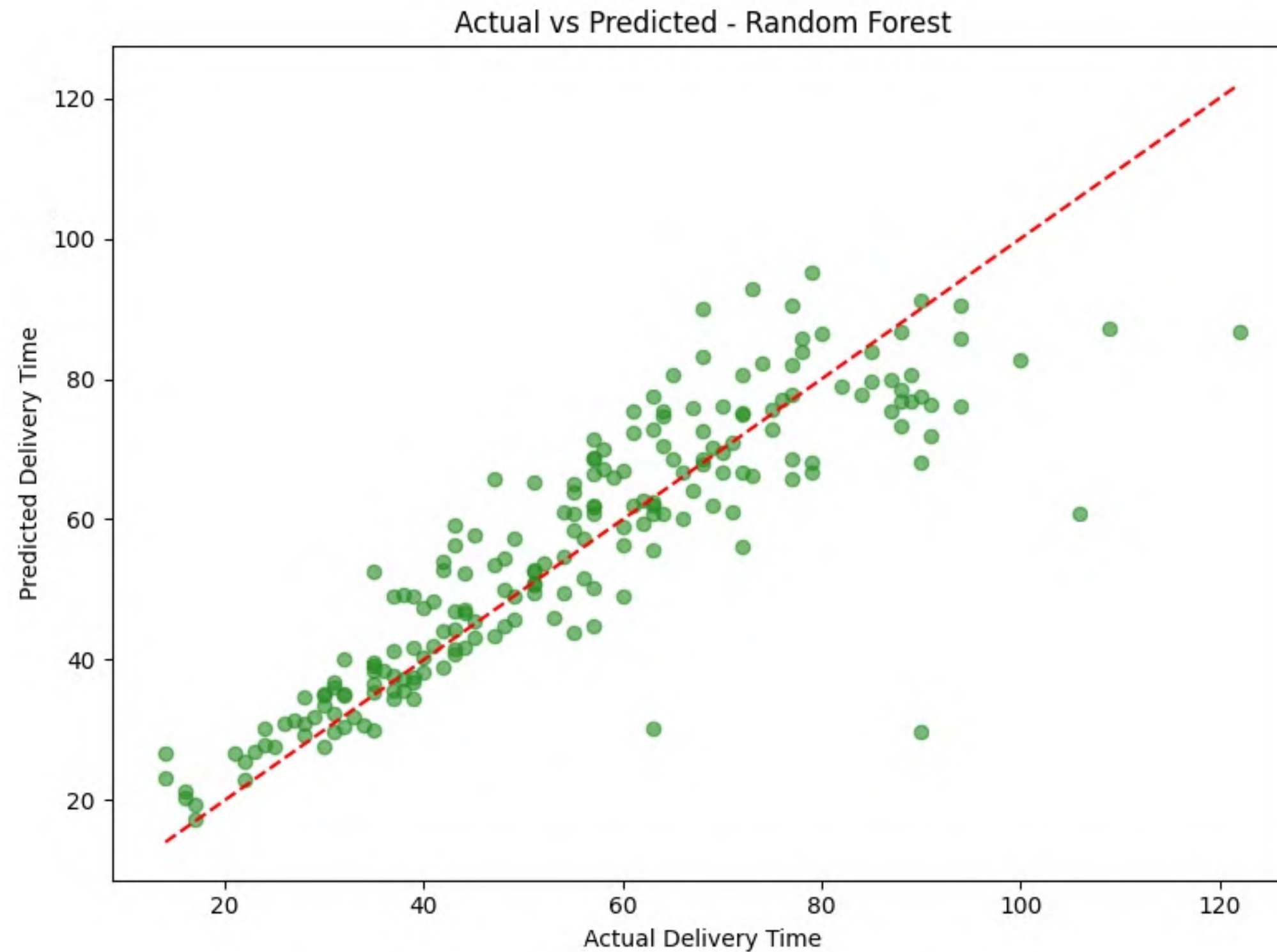




REGRESSION COMPARISON

- Linear Regression, Ridge Regression, Lasso Regression
- Previously, LG has the lowest *MAE*. Need to see the comparison between the family.
- Basically, almost no difference.
- We keep continue using Linear Regression for tuning

MAE (minutes) — lower is better
Best MAE: Linear Regression (5.90 min)

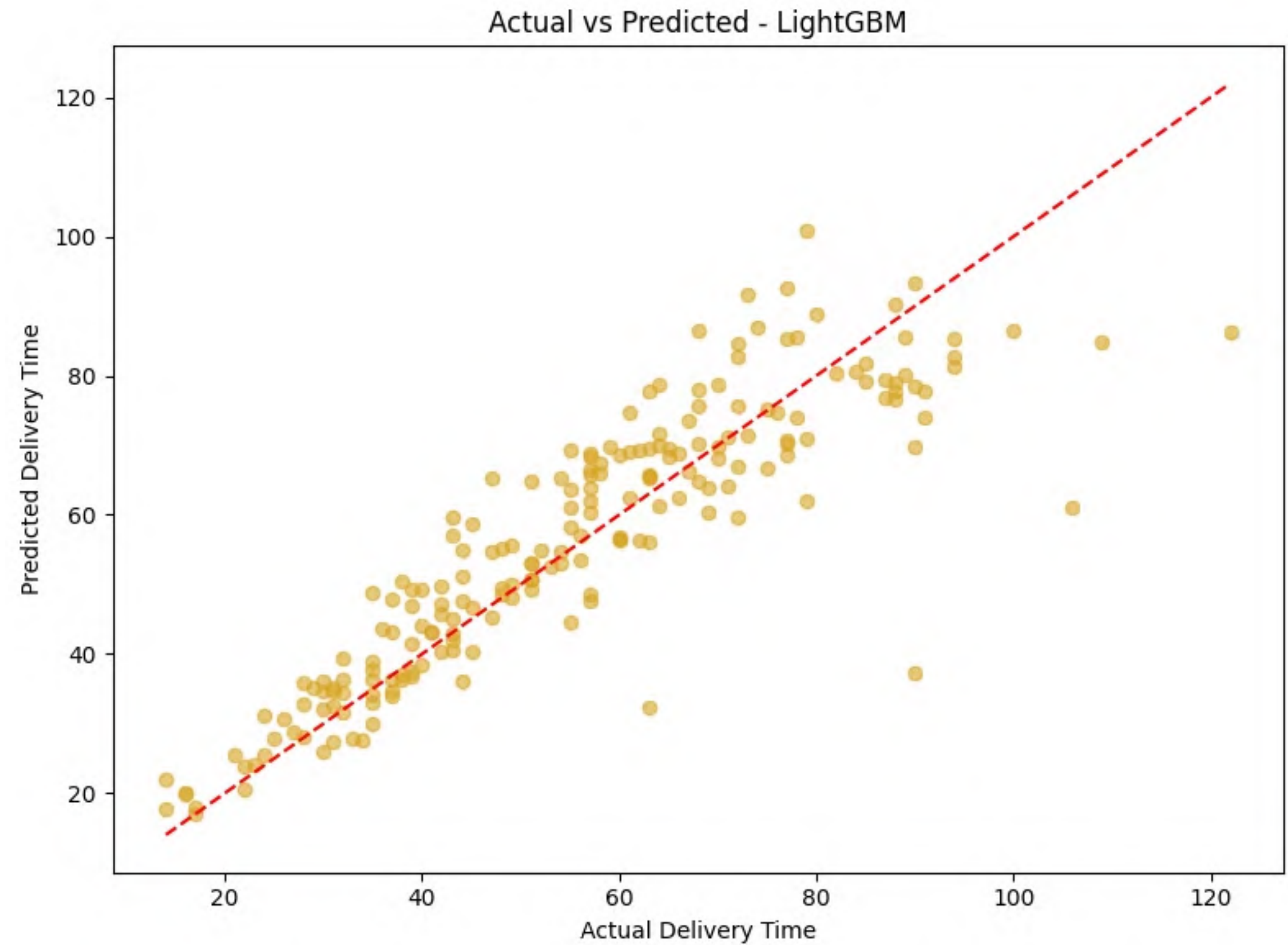


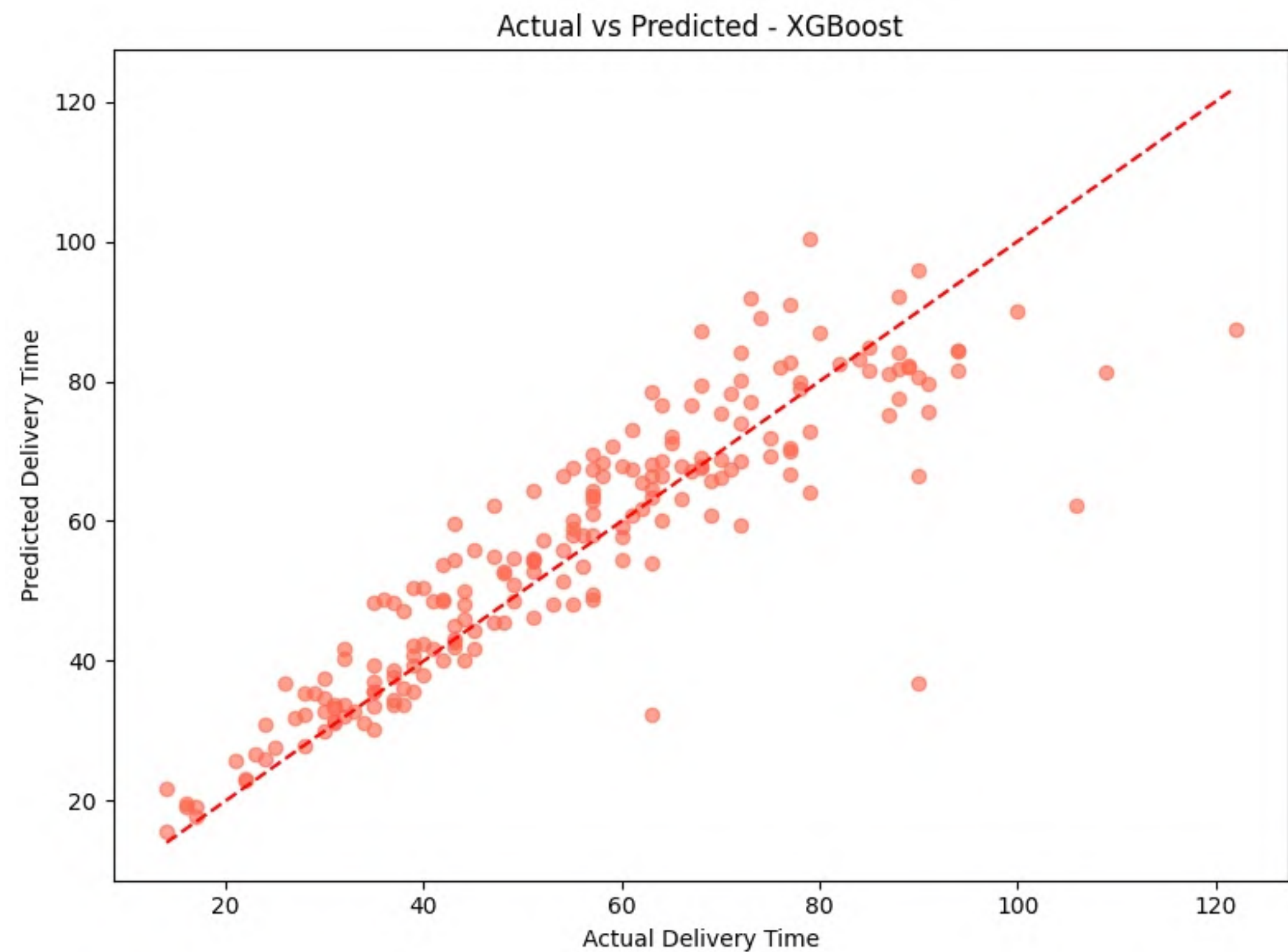
RANDOM FOREST HYPERPARAMETER TUNING

- ✓ Tuned RF MAE : 6.98. Best the untuned
- ✓ Best Parameters: {'max_depth': None, 'min_samples_leaf': 2, 'min_samples_split': 5, 'n_estimators': 200}
- ✓ errors increase for long deliveries.
- ✓ Often because very long deliveries are rare in the training data.
- ✓ Missing interactions/features that explain extreme delays

LIGHTGBM REGRESSOR TUNING

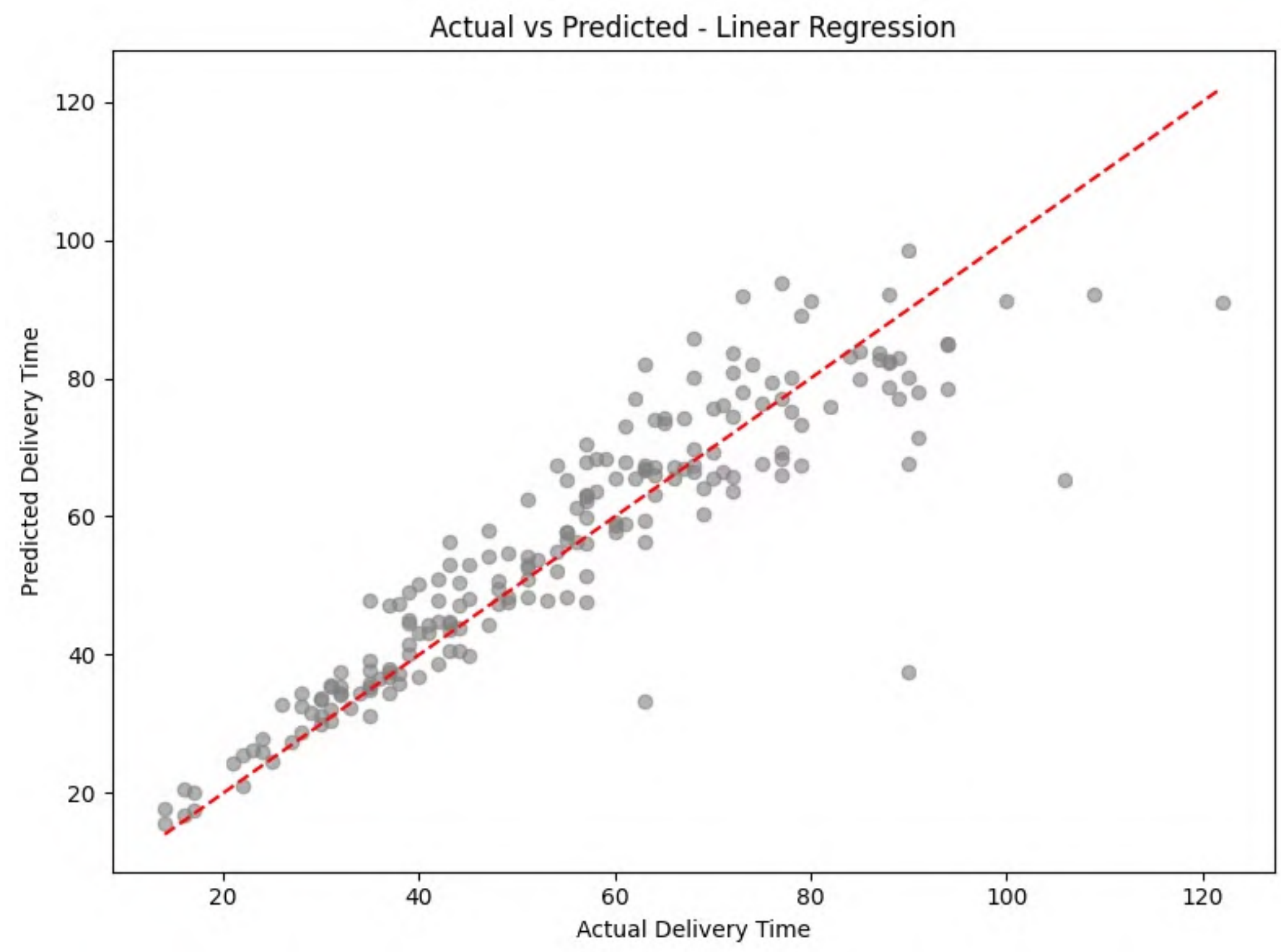
- ✓ Tuned RF MAE : 6.85. Best the untuned
- ✓ Best Parameters: {'learning_rate': 0.05, 'max_depth': 10, 'n_estimators': 100, 'num_leaves': 31}
- ✓ Same as RF, errors increase for long deliveries.
- ✓ The gain is minor





XGBOOST TUNING

- ✓ Tuned RF MAE : 6.46. Best the untuned
- ✓ Best Parameters: {'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 100, 'subsample': 0.8}
- ✓ Same as RF, errors increase for long deliveries.
- ✓ The gain is minor

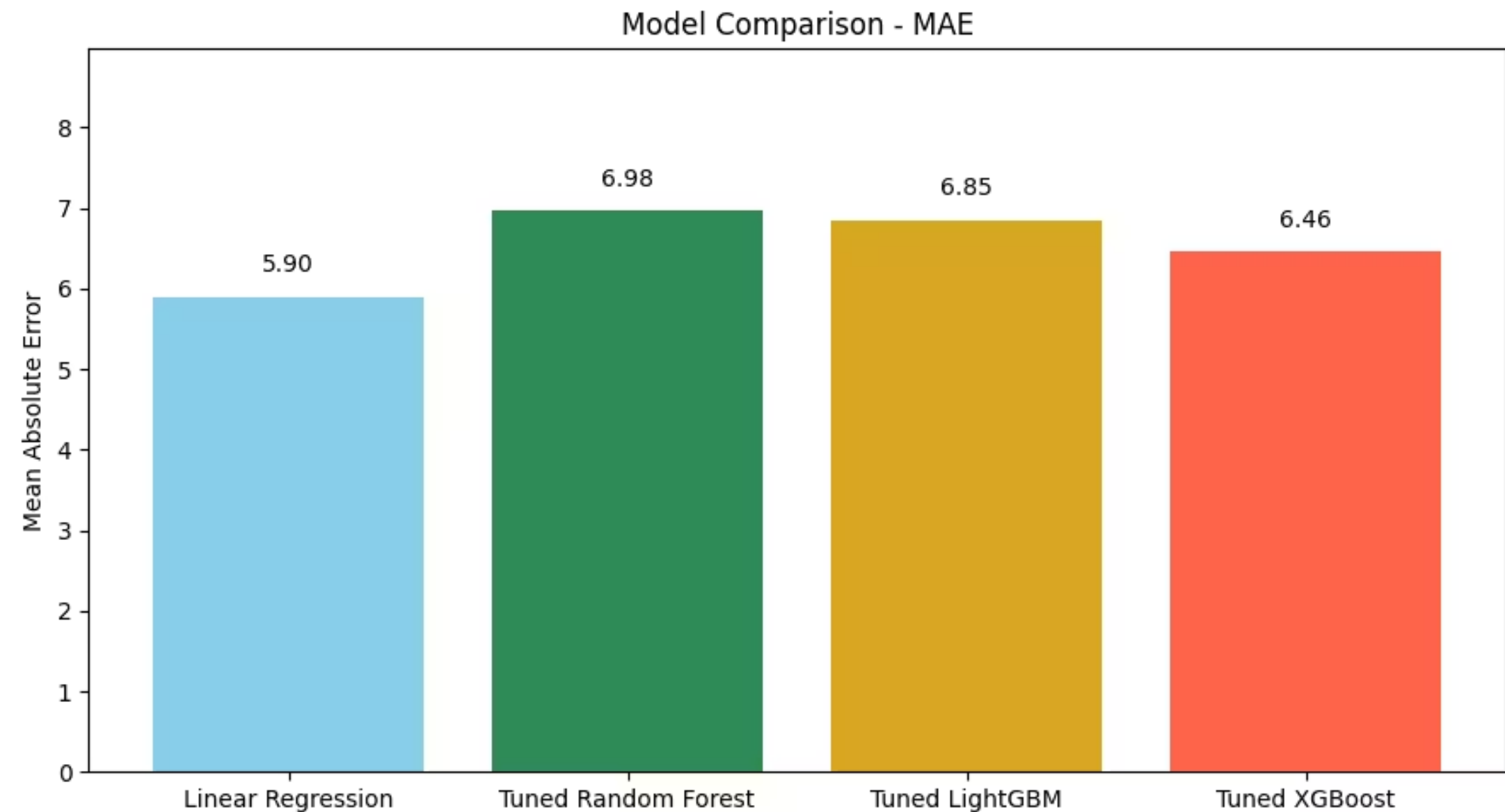


LINEAR REGRESSION

- ✓ The scatter plot looks more hugging to the actual or to the perfect predictions

MAE MODEL COMPARISON

- ✓ Linear regression still bests all other models.
- ✓ Linear regression has only 5.90 MAE
- ✓ Despite experimenting with various boosting and ensemble techniques, the simplicity and effectiveness of Linear Regression prevailed, suggesting a strong linear relationship between the features and the target variable.
- ✓ Therefore, Linear Regression is selected as the final model for deployment.



BUSINESS RECOMMENDATION

- ✓ Micro zoning/batching (reduce distance)
- ✓ Create ETA buffer for a different weather
- ✓ Avoid traffic congestion
- ✓ Experience driver, long distance delivery





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